

Tool: Migration & Schema QC Checklist

Apa: daftar periksa pre-handoff — gabungan QC schema design (SOP-SH-01) + migration safety (SOP-SH-02). Hard gate sebelum skema/migration lanjut ke Pre-Tata Gate. **Kapan dipake:** tiap mau handoff skema baru atau migration. **Kenapa:** **DOWN yang gak teruji = belum lolos.** Data SACRED enforce di sini (kontrol D1–D6). **Framework:** COBIT APO14/BAI03, mandate Data SACRED.

CHECKLIST (semua wajib sebelum handoff)

Schema (SOP-SH-01)

- ☐ **3NF default**, denormalisasi (kalau ada) ada justifikasi tertulis
- ☐ **PK defined** (UUID v7 / serial)
- ☐ **FK constraints** — integritas referensial, no orphan
- ☐ **NOT NULL** untuk field wajib; NULL hanya buat "unknown"
- ☐ **UNIQUE** untuk natural key
- ☐ **CHECK** untuk enum/range (status pakai CHECK, bukan free-text)
- ☐ **deleted_at** ada (soft-delete) — Data SACRED (kontrol D1)
- ☐ **Audit cols** lengkap: `created_at` / `updated_at` / `created_by` / `updated_by` (kontrol D2)
- ☐ **Ledger uang isolated** kalau ada uang — append-only, terpisah customer (kontrol D6)

Indexes (SOP-SH-05)

- ☐ **Index per query pattern** — dibuktikan EXPLAIN (kontrol D5)
- ☐ **No redundant index** — `(a,b)` udah cover `(a)`
- ☐ **Partial index** buat filter soft-delete (`WHERE deleted_at IS NULL`)

Migration (SOP-SH-02)

- ☐ **UP script** ada
- ☐ **DOWN script** ada **DAN teruji** di staging (benar-benar balik ke kondisi awal) — kontrol D3
- ☐ **Reversibility diklasifikasi** — destruktif = Type-1  escalate Tata
- ☐ **Lock duration assessed** — tabel besar pakai online DDL (`pg_repack/pt-osc`), index `CONCURRENTLY`
- ☐ **Backfill chunked + idempotent** kalau tabel besar
- ☐ **Tested di staging** dengan data volume realistic
- ☐ **No hard delete / overwrite** tanpa izin Tata (Data SACRED)

Multi-tenant (kalau applicable)

- ☐ **Tenant isolation** — row-level security atau app-level filter (`tenant_id`)

Performance (SOP-SH-03)

- [] **EXPLAIN ANALYZE done** untuk query kritikal
- [] **No seq scan** di tabel besar pada jalur panas

PDP / Keamanan (SOP-SH-04)

- [] **Data pribadi terklasifikasi** + aturan retensi (kontrol D4)
- [] **Data pribadi spesifik** dienkripsi at-rest (ISO 27001)

Handoff

- [] **Joint review @saitama** — BE setuju shape data
- [] (Type-1) **Schema Decision Record** ada + ter-escalate

CONTOH TERISI (Proyek-Contoh — migration: tambah kolom **hp** ke **rsvp**)

Konteks: produk minta no HP tamu buat reminder. Tabel rsvp existing ~5rb row.

- [x] Reversibility: ADD COLUMN nullable = Type-2 (reversible) → ● no escalate
- [x] UP: ALTER TABLE rsvp ADD COLUMN hp TEXT; (nullable, no backfill, no lock lama)
- [x] DOWN: ALTER TABLE rsvp DROP COLUMN hp; (teruji di staging: kolom balik hilang, data lain
- [x] Lock duration: ADD nullable column = metadata-only di PG11+, instant, no rewrite ✓
- [x] No hard delete / overwrite ✓
- [x] PDP: hp = data pribadi → ditambah ke data-classification-pdp + aturan retensi ✓
- [x] Joint review @saitama: BE setuju, hp masuk DTO RSVP ✓

Verdict: LOLOS. Type-2, aman, reversible, DOWN teruji. Deploy via @levi (no window khusus).

*Catatan: kalau requirement-nya **hp NOT NULL**, JANGAN langsung — itu lock full-table rewrite + gagal di baris lama yang null. Caranya: add nullable → backfill chunked → baru set NOT NULL. Beda 3 langkah, beda nasib produksi.*